통계열역학 (2010년 1학기) 기말고사 해답

(총 125점)

1. **(총 20점, 각 4점)**

   (a) 6. 축퇴도

   (b) 14. Maxwell-Boltzman 분포

   (c) 2. 정준 앙상블

   (d) 5. 등분배 원리 (모두 정답 인정)

   (e) 12. 대칭수

2. **(총 20 점, 각 10 점)**

   (a) 에너지 요동 σ_E^2 은

$$\begin{aligned}\sigma_E^2 &= \langle E - \langle E\rangle\rangle, \\ &= \langle E^2\rangle - \langle E\rangle^2,\end{aligned}$$

   한편, $\langle E^2\rangle$ 는

$$\begin{aligned}\langle E^2\rangle &= \sum_j E_j^2 p_j = \frac{1}{Q}\sum_j E_j^2 e^{-\beta E_j}, \\ &= -\frac{1}{Q}\frac{\partial}{\partial\beta}\left(\sum_j E_j e^{-\beta E_j}\right)_{N,V} = -\frac{1}{Q}\frac{\partial}{\partial\beta}\big(\langle E\rangle Q\big), \\ &= -\frac{\partial\langle E\rangle}{\partial\beta} - \langle E\rangle\frac{\partial \ln Q}{\partial\beta}, \\ &= -\frac{\partial\langle E\rangle}{\partial\beta} + \langle E\rangle\langle E\rangle, \\ &= k_B T^2\frac{\partial\langle E\rangle}{\partial T} + \langle E\rangle^2.\end{aligned}$$

   그러므로,

$$\sigma_E^2 = k_B T^2\left(\frac{\partial\langle E\rangle}{\partial T}\right)_{N,V} = k_B T^2 C_V.$$

   (b) 단원자분자의 평균에너지는 $\langle E\rangle = \frac{3}{2}Nk_BT$ 이므로 $C_V = \partial E/\partial T = \frac{3}{2}Nk_B$ 이다. 그러므로,

$$\frac{\sigma_E}{\langle E\rangle} = \frac{\sqrt{k_B T^2 C_V}}{\langle E\rangle} = \frac{\sqrt{3/2N}k_BT}{3/2Nk_BT} = \frac{1}{\sqrt{3/2N}}.$$

   만약 $N\to\infty$,

$$\lim_{N\to\infty}\frac{\sigma_E}{\langle E\rangle} = \lim_{N\to\infty}\frac{1}{\sqrt{3/2N}} = 0$$

3. **(총 10 점, 각 5 점)**

(a) 평균에너지 $\langle E\rangle$ 는

$$\begin{aligned}
\langle E\rangle &= k_BT^2\left(\frac{\partial \ln Q}{\partial T}\right)_{N,V} \\
&= k_BT^2\frac{\partial}{\partial T}\ln\left[\frac{1}{N!}\left(\frac{2\pi mk_BT}{h^2}\right)^{3N/2}(V-Nb)^N e^{aN^2/(Vk_BT)}\right] \\
&= k_BT^2\frac{\partial}{\partial T}\left\{-\ln N! + \frac{3N}{2}\ln\left(\frac{2\pi mk_B}{h^2}\right) + \frac{3N}{2}\ln T + N\ln(V-Nb) + \frac{aN^2}{Vk_BT}\right\} \\
&= \frac{3}{2}Nk_BT - \frac{aN^2}{V}
\end{aligned}$$

C_V 는

$$C_V = \left(\frac{\partial\langle E\rangle}{\partial T}\right)_{N,V} = \frac{3}{2}Nk_B$$

(b)

$$\begin{aligned}
P &= -\left(\frac{\partial A}{\partial V}\right)_{N,T} \\
&= k_BT\left(\frac{\partial \ln Q}{\partial V}\right)_{N,T} \\
&= k_BT\frac{\partial}{\partial V}\left\{\ln\left[\frac{1}{N!}\left(\frac{2\pi mk_BT}{h^2}\right)^{3N/2}(V-Nb)^N e^{aN^2/(Vk_BT)}\right]\right\} \\
&= k_BT\frac{\partial}{\partial V}\left\{-\ln N! + \frac{3N}{2}\ln\left(\frac{2\pi mk_BT}{h^2}\right) + N\ln(V-Nb) + \frac{aN^2}{Vk_BT}\right\} \\
&= k_BT\left\{\frac{N}{V-Nb} - \frac{aN^2}{V^2k_BT}\right\} \\
&= \frac{Nk_BT}{V-Nb} - \frac{aN^2}{V^2}
\end{aligned}$$

4. **(총 20 점, 각 10 점)**

(a) 평균에너지 $\langle E\rangle$ 는

$$\begin{aligned}
\langle E\rangle &= k_BT^2\left(\frac{\partial \ln Q}{\partial T}\right)_{N,S} \\
&= k_BT^2\frac{\partial}{\partial T}\ln\left[\frac{1}{N!}\left(\frac{2\pi mk_BT}{h^2}\right)^{N}S^N\right] \\
&= k_BT^2\frac{\partial}{\partial T}\left\{-\ln N! + N\ln\left(\frac{2\pi mk_B}{h^2}\right) + N\ln T + N\ln S\right\} \\
&= Nk_BT
\end{aligned}$$

C_V 는

$$C_V = \left(\frac{\partial\langle E\rangle}{\partial T}\right)_{N,V} = Nk_B$$

(b)

$$\begin{aligned}
\phi &= -\left(\frac{\partial A}{\partial S}\right)_{N,T} \\
&= k_B T\left(\frac{\partial \ln Q}{\partial S}\right)_{N,T} \\
&= k_B T \frac{\partial}{\partial S}\left\{ \ln\left[\frac{1}{N!}\left(\frac{2\pi m k_B T}{h^2}\right)^N S^N\right]\right\} \\
&= k_B T \frac{\partial}{\partial S}\left\{ -\ln N! + N\ln\left(\frac{2\pi m k_B T}{h^2}\right) + N\ln S\right\} \\
&= \frac{N k_B T}{S}
\end{aligned}$$

5. **(총 20 점, 각 5 점)**

(a)

$$\begin{aligned}
q_{elec} &= g_0 e^{-\epsilon_0/k_B T} + g_1 e^{-\epsilon_1/k_B T} \\
&= g_0 + g_1 e^{-\epsilon_1/k_B T} \qquad (\epsilon_0 = 0 \text{ 이므로})
\end{aligned}$$

(b)

$$Q = (g_0 + g_1 e^{-\epsilon_1/k_B T})^N$$

(c)

$$\begin{aligned}
\langle E\rangle &= k_B T^2\left(\frac{\partial \ln Q}{\partial T}\right)_{N,V} \\
&= k_B T^2 \frac{\partial}{\partial T}\left[\ln\left(g_0 + g_1 e^{-\epsilon_1/k_B T}\right)^N\right] \\
&= k_B T^2 N \frac{g_1 \epsilon_1 e^{-\epsilon_1/k_B T}}{k_B T^2}\frac{1}{g_0 + g_1 e^{-\epsilon_1/k_B T}} \\
&= \frac{N g_1 \epsilon_1 e^{-\epsilon_1/k_B T}}{g_0 + g_1 e^{-\epsilon_1/k_B T}}
\end{aligned}$$

(d) i. $T \to 0$ 일 때,

$$\begin{aligned}
\lim_{T\to 0}\langle E\rangle &= \lim_{T\to 0}\frac{N g_1 \epsilon_1 e^{-\epsilon_1/k_B T}}{g_0 + g_1 e^{-\epsilon_1/k_B T}} \\
&= 0 \qquad (\lim_{T\to 0} e^{-\epsilon_1/k_B T} = 0 \text{ 이므로})
\end{aligned}$$

ii. $T \to \infty$ 일 때,

$$\begin{aligned}\lim_{T\to\infty}\langle E\rangle &= \lim_{T\to\infty}\frac{Ng_1\epsilon_1 e^{-\epsilon_1/k_BT}}{g_0+g_1e^{-\epsilon_1/k_BT}} \\ &= \lim_{T\to\infty}\frac{Ng_1\epsilon_1}{g_0+g_1} \qquad (\lim_{T\to\infty} e^{-\epsilon_1/k_BT}=1 \text{ 이므로})\end{aligned}$$

6. **(총 10 점, 각 5 점)**

(a) $\Theta_{rot}(\mathrm{HCl}) = 15.02$ K 이므로 1000 K 에서 $\Theta_{rot}/T = 15.02/1000 = 0.01502$ 이다 그러므로 1000 K 에서 HCl 분자가 $J=4$ 에 존재할 확률은 다음과 같이 주어진다.

$$p_{J=4} = \frac{\Theta_{rot}}{T}(2J+1)e^{-J(J+1)\Theta_{rot}/T} = (0.01502)(2\times 4+1)e^{-0.01502\times(4\times5)} = 0.100$$

(b) $\Theta_{vib}(\mathrm{HCl}) = 4227$ K 이므로 $\Theta_{vib}/T = 4227/1000 = 4.227$ 이다. 그러므로 1000 K 에서 HCl 분자가 $v=2$ 에서 존재할 확률은 다음과 같이 주어진다.

$$\begin{aligned}p_v(T) &= \frac{e^{-\beta\epsilon}}{q} \\ &= \frac{e^{-\beta(v+1/2)h\nu}}{e^{-\beta h\nu/2}/(1-e^{-\beta h\nu})} \\ &= e^{-\beta h\nu v}\left(1-e^{-\beta h\nu}\right) \\ &= e^{-v\Theta_{vib}/T}\left(1-e^{-\Theta_{vib}/T}\right) \\ &= e^{-4.227v}\left(1-e^{-4.227}\right) \\ &= e^{-4.227\times2}\left(1-e^{-4.227}\right) = 2.099\times10^{-4}\end{aligned}$$